SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 6

Decision Trees

Decision trees visualise sequential decisions under uncertainty: decision and chance nodes, probabilities, payoffs. Build and roll back with expected monetary value, multi-stage decisions, EVPI, risk attitude and utility, and probability sensitivity.

Material

1 Elements and construction

Decision nodes (□), chance nodes (○, probabilities sum to 1), payoffs. Draw left to right in time; roll back right to left: EMV = Σ p × payoff at chance nodes, choose the best branch at decision nodes.

2 The example tree, redrawn Deep dive

Decision	Chance	Payoff	EMV
Big	High 0.6	10	4.4 ✓
	Low 0.4	−4	
Small ‖	High 0.6	5	3.4
	Low 0.4	1	
None ‖	—	0	0

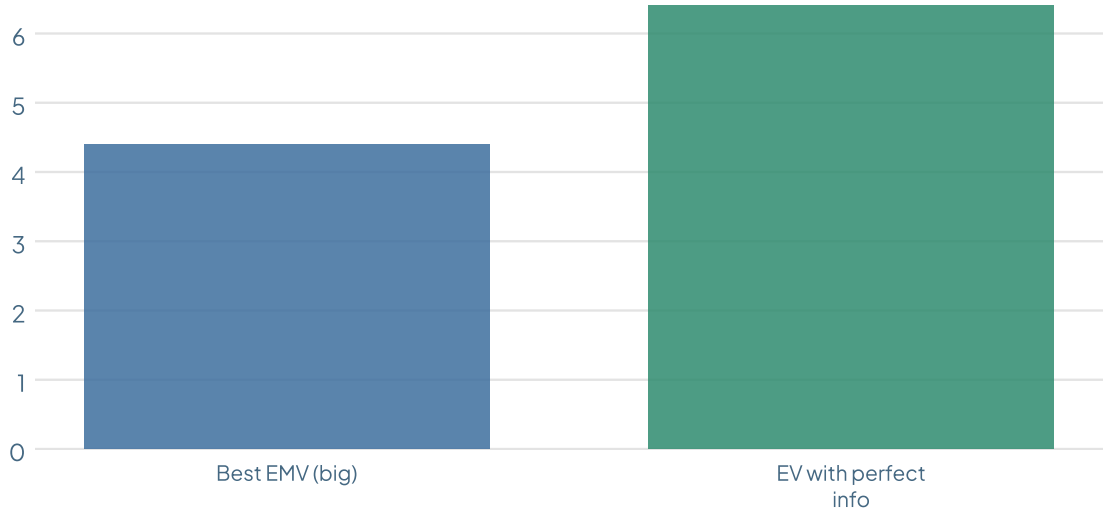
3 Worked example

Example: launch big EMV 4.4, small 3.4, none 0 → big. EVPI = 6.4 – 4.4 = 2.0. Indifference probability p = 0.5. Staged decisions keep valuable options open. The class example (power design, 6.75 vs 6.2) follows the same logic.

4 Simulator: EMV, indifference probability and EVPI Deep dive

The lines cross at p = 0.5.

Expected value: without vs with perfect information



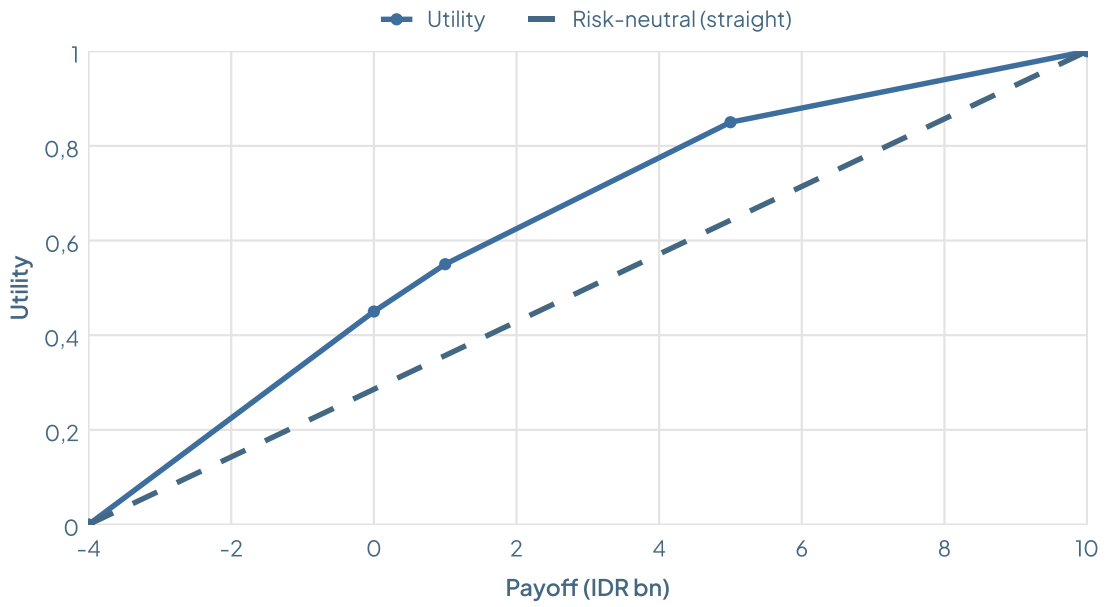
EVPI = 2.0: the ceiling on research spend.

5 Risk, utility and limitations

EMV assumes risk neutrality; risk-averse decision makers use utilities (probability-equivalence) and expected utility, and examine risk profiles. Trees grow bushy, probabilities are bias-prone, and one-off decisions differ from long-run averages — always test sensitivity.

6 Utility: why people reject the highest EMV Deep dive

Risk-averse utility function (illustrative)




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EU big = 0.60 · EU small = 0.73
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With a concave (risk-averse) utility, small wins despite a lower EMV.

7 Imperfect information: valuing market research (EVSI) with Bayes

EVPI (IDR 2bn in the product-launch example) is an upper bound. Real research is imperfect, so we need the **expected value of sample information (EVSI)**.

Data: prior P(high) = 0.6, P(low) = 0.4. The survey's reliability: if demand is truly high, the survey says "positive" 80% of the time; if low, it still says "positive" 30% of the time.

Step	Calculation	Result
P(positive)	0.6×0.8 + 0.4×0.3 = 0.48 + 0.12	0.60
P(negative)	1 − 0.60	0.40
P(high | positive)	0.48 ÷ 0.60	0.80

P(high | negative)	0.6×0.2 ÷ 0.40 = 0.12 ÷ 0.40	0.30

Rollback after the survey result (IDR bn):

Survey result	EMV large scale	EMV small scale	Choose
Positive (0.60)	0.8×10 + 0.2×(−4) = **7.2**	0.8×5 + 0.2×1 = 4.2	Large → 7.2
Negative (0.40)	0.3×10 + 0.7×(−4) = 0.2	0.3×5 + 0.7×1 = **2.2**	Small → 2.2

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EV with survey = 0.60×7.2 + 0.40×2.2 = 5.2
EVSI = 5.2 - 4.4 = 0.8  ·  Efficiency = EVSI ÷ EVPI = 0.8 ÷ 2.0 = 40%
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How to read it

The survey is worth buying only if it costs less than **IDR 0.8bn**. It has value because it *can change the decision*: without it we always choose large scale; with it, a negative result moves us to small scale. Information that cannot change the decision is worth zero, however expensive the research.

Risk profile. Besides EMV, draw the distribution of outcomes for each strategy. Large scale without a survey: 60% chance of +10, 40% chance of −4. The "survey first" strategy cuts the chance of the −4 loss to 0.60×0.2 = just 12% (a loss only happens if the survey is positive but demand turns out low). For a risk-averse firm, this lower chance of loss may matter more than the higher EMV.

Key takeaways

1. Decision, chance, payoff nodes.
2. Roll back: EMV = Σ p × payoff.
3. EVPI.
4. Indifference probabilities.
5. Expected utility for risk aversion.

Case studies

Case & worked answers

Linking to Pelindo (MK4) and BUMDes

Use the tools to quantify your Pelindo tree (probabilities for demand and subsidence, NPV payoffs, EMV, EVPI of a geotechnical study). SMART fits many criteria with little uncertainty; trees fit uncertain events — they can be combined.

Worked examples

Problem 1 · Medium · **EMV, EVPI, dan titik impas probabilitas**

Investasi produk baru: sukses p = 0,3 → +Rp50 M; gagal → –Rp10 M. Tidak investasi = 0. (a) Keputusan EMV? (b) Berapa maksimal membayar riset sempurna? (c) Di p berapa investasi mulai layak?

1. (a) EMV investasi = 0,3 × 50 + 0,7 × (–10) = 15 – 7 = **8** > 0 → investasi.
2. (b) Dengan info sempurna: jika sukses, investasi (50); jika gagal, tidak (0). EV = 0,3 × 50 + 0,7 × 0 = 15.
3. EVPI = 15 – 8 = **7**.
4. (c) Impas ketika 50p – 10(1 – p) = 0 → 60p = 10 → p = 0,167.

Answer: Investasi (EMV Rp8 M); riset maksimal Rp7 M; layak selama p > 16,7%.

Practice questions

Try answering first. The answer key is below each question.

1. At a chance node you compute…

A. Max payoff
B. Σ p × payoff
C. Min payoff
D. Simple mean

Answer: B. EMV.

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2. EVPI is 2.0. Research costing 2.5 is…

A. Worth it
B. Not worth it
C. Always worth it
D. Can't tell

Answer: B. Imperfect research is worth even less.

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3. Rejecting the top EMV for fear of big loss is…

A. Risk-seeking
B. Risk-neutral
C. Risk-averse
D. Irrational

Answer: C. Concave utility.

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4. Big and small are equal when p(high) = …

A. 0.4
B. 0.5
C. 0.6
D. 0.7

Answer: B. $14p - 4 = 4p + 1$.

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5. Trees are drawn left to right following…

A. Payoff size
B. Time order
C. Alphabet
D. Probability

Answer: B. Solved in reverse.

Sources

- [Complete] 06 Decision Tree (MK5) — slide berupa gambar; contoh di sini disusun ulang
- Goodwin & Wright, Bab 5–6